

Classification of LV fuses

LECTURE 3

Choosing the Right Fuse Size

- While selecting the proper fuse and its rated size for electrical appliances is based on different factors and environments. But the following basic formula shows **how to choose the right size of fuse?**
- Selecting the proper fuse and its rated size for electrical appliances depends on various factors and environmental conditions. However, the following basic formula can guide you in **selecting the right fuse size for typical applications:**

$$\text{Fuse Rating} = (\text{Power} \div \text{Voltage}) \times 1.25$$

To determine the proper fuse size for an electrical appliance, consider the following example: For an outlet with a total load of 480 W connected to a 120V supply, you can calculate the required fuse rating.

For example:

$$\text{Fuse Rating (in Amps)} = (\text{Load in Watts} / \text{Supply Voltage}) \times 1.25$$
$$(480\text{W} \div 120\text{V}) \times 1.25 = \mathbf{5A}$$

This formula can be applied to 230V or 240V circuits as well.

In this case of 120V circuit, the minimum fuse size should be equal to 5 Amps to or higher than the calculated value.

In the above case, the minimum size of the fuse should not be less than 5A for safe and reliable operation of the circuit.

Fusing factor

- The fusing factor is the smallest current that will cause the fuse element to melt.
- The fusing factor is the ratio of a fuse's minimum fusing current. This may vary between 1.25 and 2.5 times the current rating.
- The current rating is the nominal rated current in Amps marked on the fuse body that the fuse will carry continuously without weakening.

$$\text{Fusing factor} = \frac{\text{Rated minimum fusing current}}{\text{Current rating}}$$

$$\text{Rated minimum fusing current} = \text{Fusing factor} * \text{Current rating}$$

Classification of LV fuses

- There are four classes, depending on fusing factors .

- **1- Class P:-** Having a fusing factor of 1.25 or less.

This class is to protect circuits from small overloads.

- **2- Class Q1 fuses:-** Used for circuits with small over-currents and higher values of overload.

Fusing factor between 1.25 and 1.5.

- **3- Class Q2 fuses:-** Used for circuits with small over-currents and higher values of overload.

Fusing factor between 1.5 and 1.75.

- **Class R fuses:** Used to protect a circuit against large over-currents only. (is mainly back-up protection).

Class R- Fusing factor between 1.75 and 2.5.

Example:

What is the minimum rated current of the 20A Q2 fuse that will be operated?

Rated minimum fusing current = Fusing factor x Current rating =
 $20\text{A} \times 1.5 = 30\text{ A}$ or $20\text{A} \times 1.75 = 35\text{A}$

Minimum fusing current rated to operate between 30A and 35 Amps.

According to IEC standard, two classes of LV cartridge fuses are widely used:

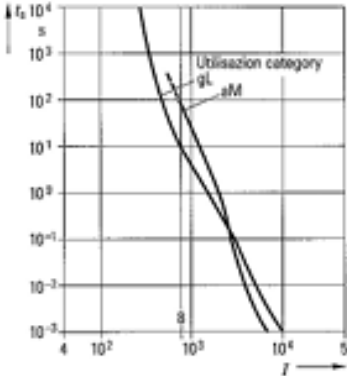
- **gG** Type:

The utilization category gG is adapted to the load characteristics of cables and wires. It is used for domestic (**for lighting circuit**) and smaller installations.

- **aM** Type

The utilization category aM is adapted to the load characteristics of motors. Overload Protection against too high warming of electrical devices at overcurrent and overload. It is used for industrial installations (for short-circuit currents). For industrial installations (**for motor protection**) **gM** type is used. "gM fuses" are industrial fuses designed for general-purpose motor protection that can withstand high motor starting surges while also providing protection for lower, continuous running currents. They are characterized by two ratings: a smaller, continuous current rating (I_n) and a larger, time/current rating (I_{ch}) indicated by an "M" (e.g., 32M63). This design allows the use of smaller-sized fuse holders and equipment for motor circuits where the continuous current is lower than the starting surge current.

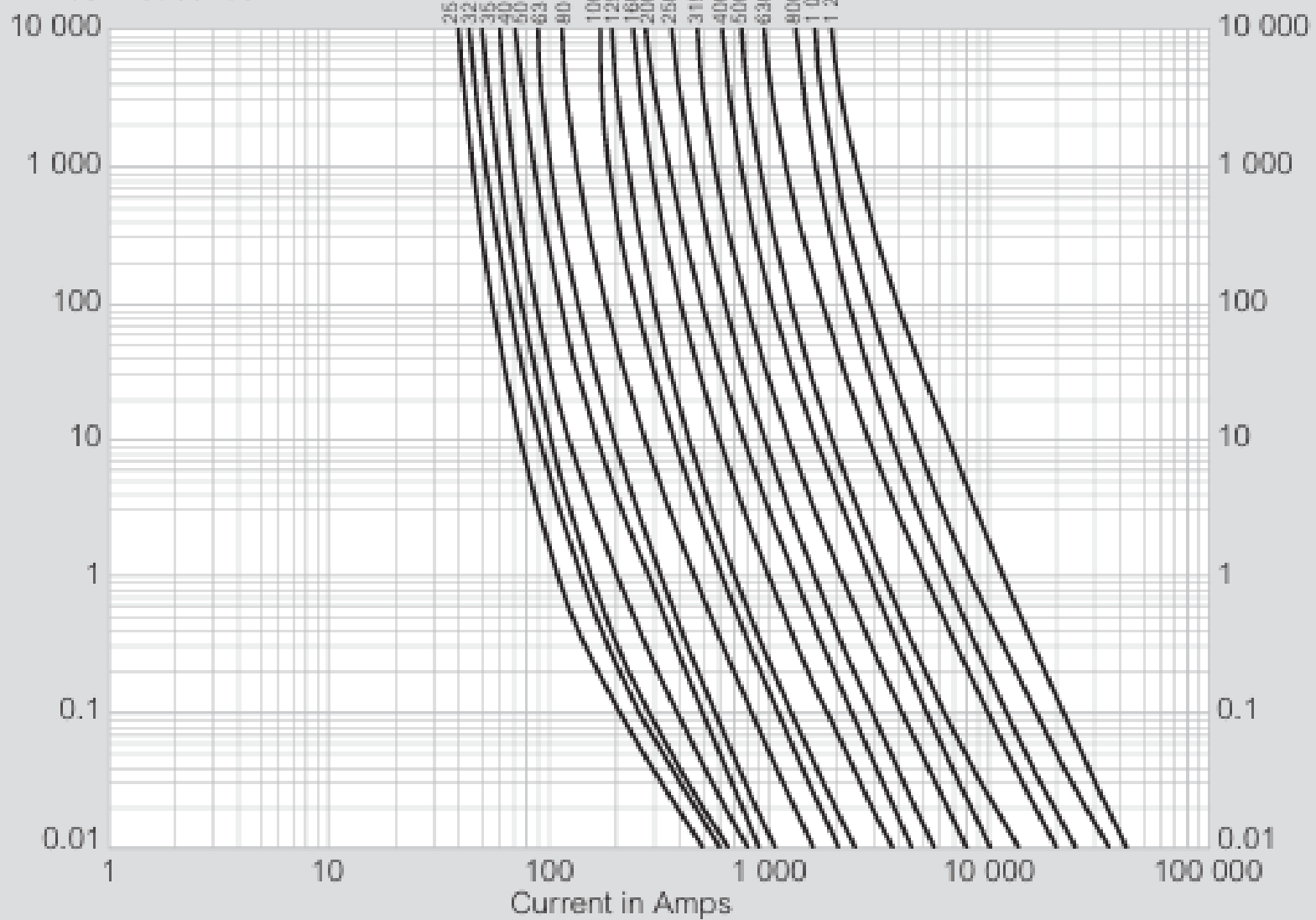
Comparison of LV HRC fuse gG - aM

Technical specifications		
	LV HRC fuse gG (3NA...)	LV HRC fuse aM (3ND...)
Standards	DIN VDE 0636-201, DIN VDE 0680-4; IEC 60269-1, -2-1; EN 60269-1; HD 630.2.1 S6	
Rated voltage	400 – 690 V AC 250 – 440 V DC	500 – 690 V AC
Rated current	2 – 1250 A	6 – 630 A
Rated breaking capacity	120 kA AC 25 kA DC	120 kA AC
Utilization category	g – full-range fuses G – general purpose	a – partial range fuses M – switchgear protection in motor circuits
Smallest rated breaking current	$\geq I_{min} (I_n)$ (smallest breaking current)	$\geq 4I_n$
Characteristics (at the same rated current)	In the upper short-circuit range the aM fuse is faster than the gG fuse. In the overload range the aM fuse is time-delayed 	
Current limitation	According standard	Better current limitation because tripping faster in upper short-circuit range.
Power dissipation (at the same rated current)	According standard	The power dissipation is less than that of the gG fuse.

Area of application		
	LV HRC fuse gG (3NA...)	LV HRC fuse aM (3ND...)
Utilization category	The utilization category gG is adapted to the load characteristics of cables and wires.	The utilization category aM is adapted to the load characteristics of motors.
Overload	Protection against too high warming of electrical devices at overcurrent and overload.	For overload protection an additional protective device is necessary. The fuse doesn't melt.
Short-circuit	Short circuit protection of switchgear, lines and cables and electrical devices	Short circuit protection of switchgear in motor circuits (also for short-circuit protection of the cable)
Rating	According the I_x of the cable or line. For motor applications overrating is necessary.	According the motor rated current.
Back-up protection	Protection of secondary circuit protective devices	
Touch voltage	Protection against the remanence of inadmissibly high touch voltages	
Universal application / example	Protection of electrical circuits, panels, motors (General purpose fuses)	Protection of load-contacts against welding in case of high overcurrents / short circuit currents

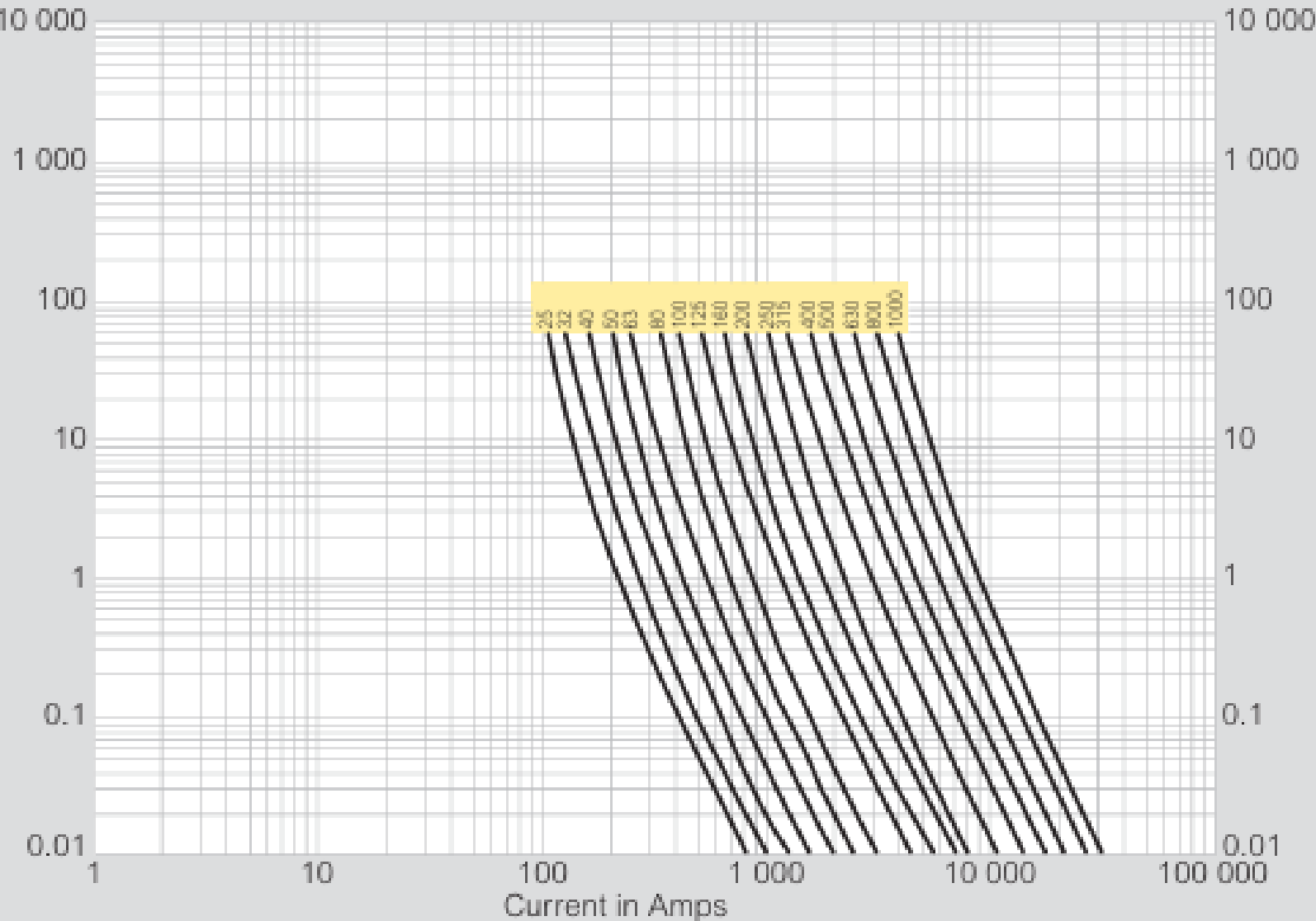
gG type

Times in seconds



aM type

Time in seconds



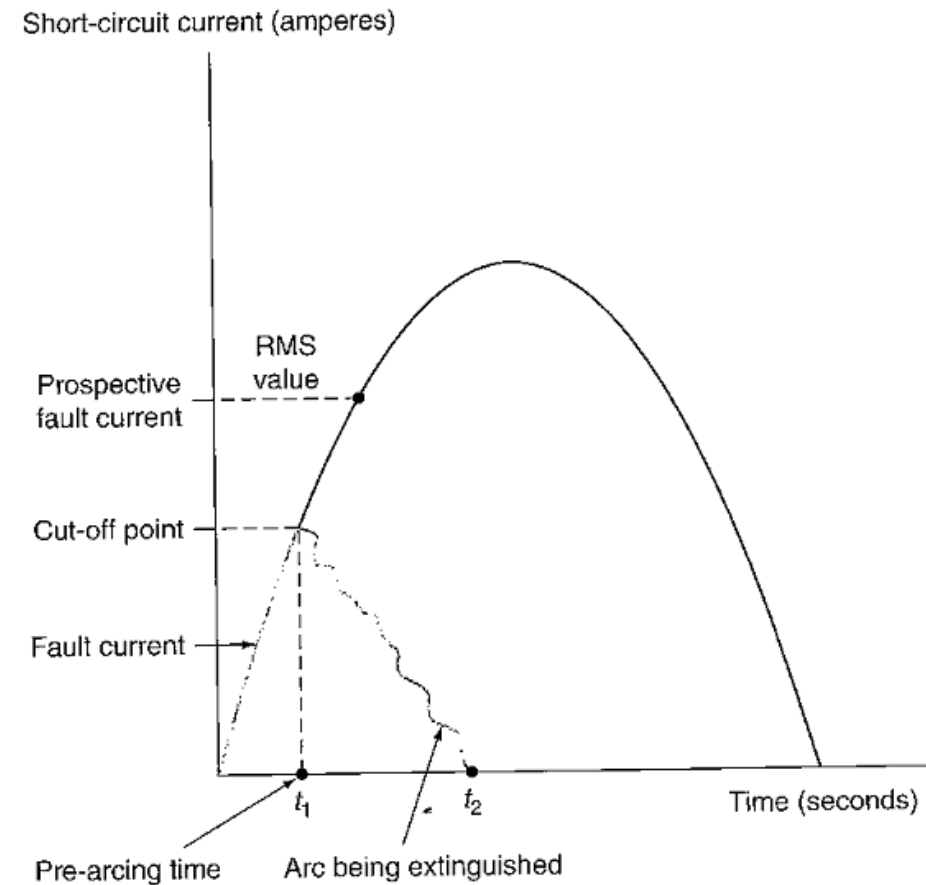
Current melting fuse element and Clearing time

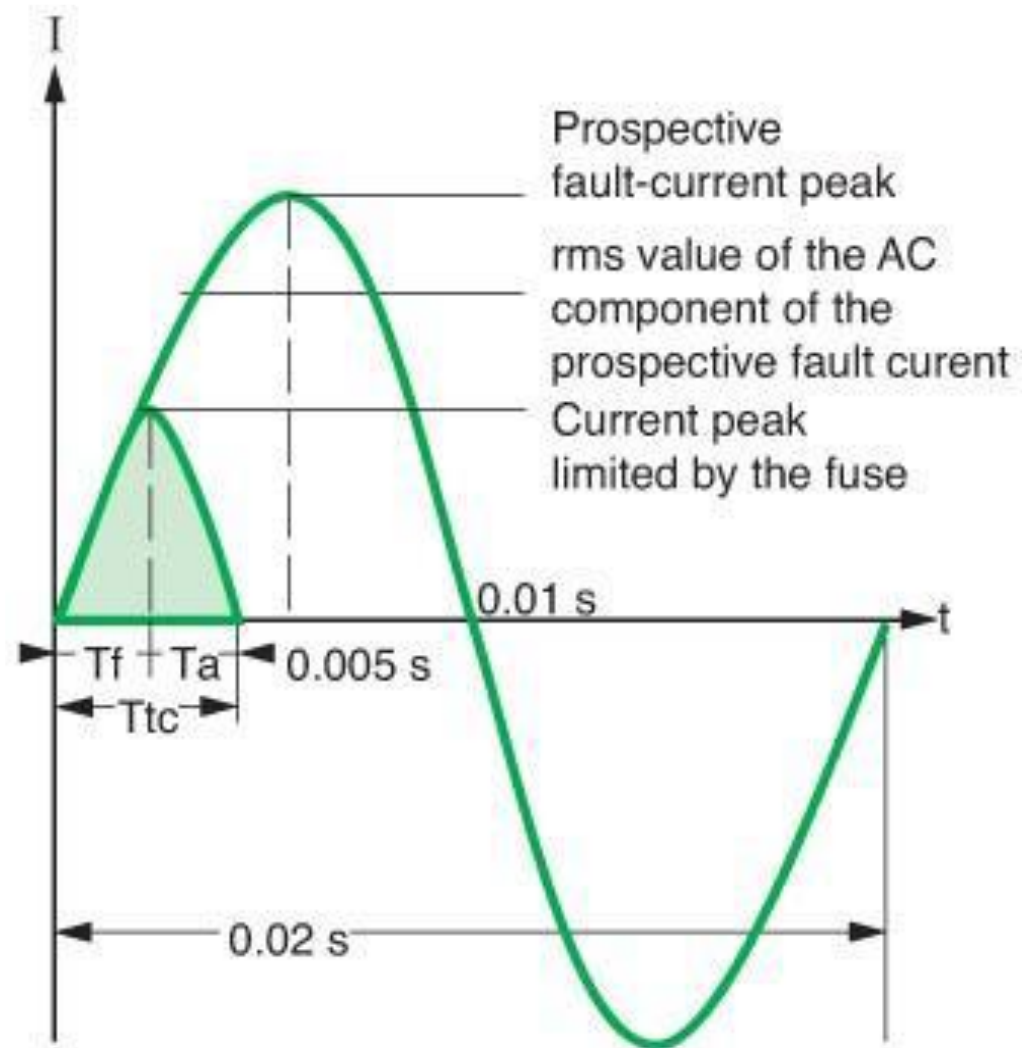
A curve to represent the total clearing time as a function of the current, with main characteristics of a fuse, the curve type may be:

- 1. The minimum melting time curve
- 2. The total clearing time curve
- 3. The fuse minimum melting current
- 4. The fuse rating (nominal current)

Short Circuit Current

- The fuse installed to protect against such current must be able to break this current as $I^2 t_1$.





T_f : Fuse pre-arc fusing time
 T_a : Arcing time
 T_{tc} : Total fault-clearance time

The I^2t factor of the fuse

- The operation time of the fuse at high levels of current is inversely proportional to the square of the current during the pre-arcing stage (operation).
- For any conductor, its temperature rise depends on the I^2t factor.
- This factor can be calculated by:
- For copper conductors:

$$I^2 t = 11.5 \times 10^4 A^2 \log_{10} \frac{273 + \theta_m}{273 + \theta_o}$$

-For Aluminum conductors:

$$I^2 t = 5.2 \times 10^4 A^2 \log_{10} \frac{273 + \theta_m}{273 + \theta_o}$$

Where:

I = Short circuit current (A).

t = Duration of the short circuit (s)

A = cross-sectional area of the conductor (mm)

θ_o = Initial temperature of the conductor (C°).

θ_m = Final temperature of the conductor (C°).

Example:

It is proposed to use a No 30 AWG (American Wire Gauge) copper wire as a fuse element for rewirable types. If its initial temperature is 50 C°, taking in consideration that the copper melts at 1083 C° calculate the following:

- (a) The $I^2 t$ needed to melt the wire.
- (b) The time needed to melt the wire if the short circuit current is 30A.

Solution

$$I^2 t = 11.5 \times 10^4 A^2 \log_{10} \frac{273 + \theta_m}{273 + \theta_o}$$

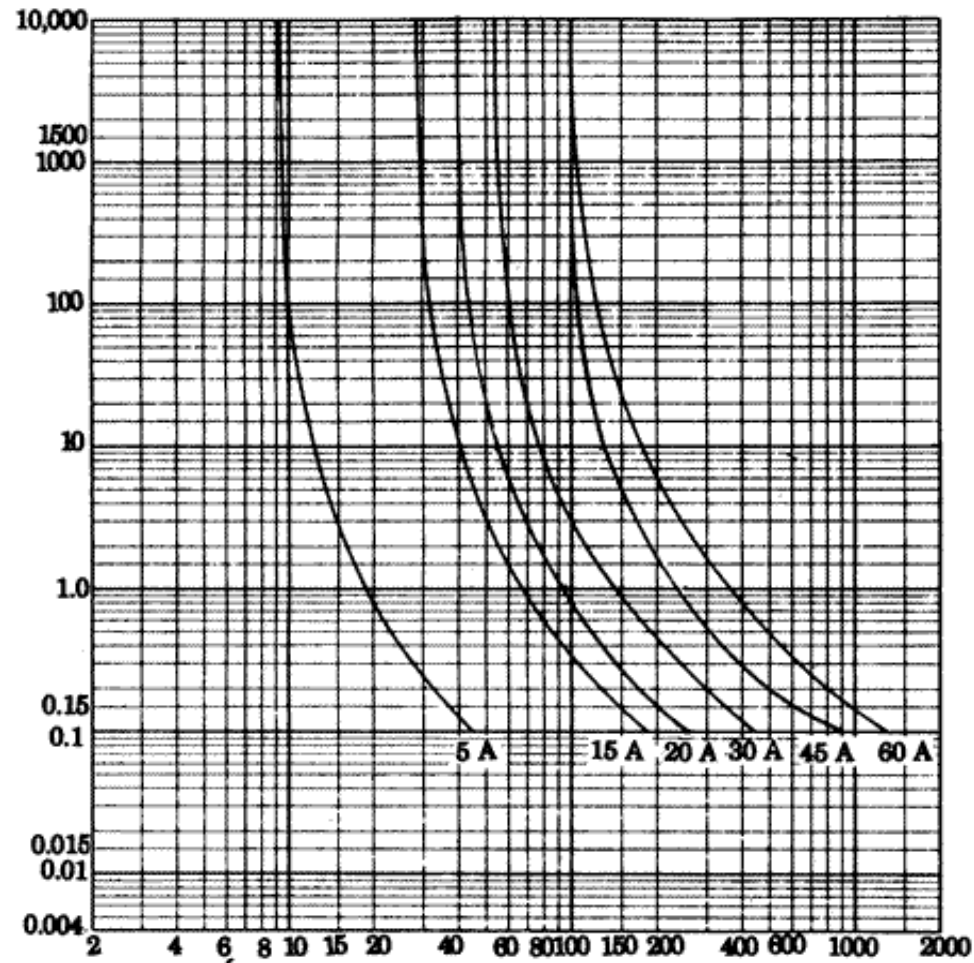
$$\begin{aligned} I^2 t &= 11.5 \times 10^4 \times 0.0503^2 \times \log_{10} \left(\frac{273 + 1083}{273 + 50} \right) = \\ &= 181.3 A^2 s \Rightarrow t = \frac{181.3}{I^2} = \frac{181.3}{30^2} = 0.2 \text{ s} \end{aligned}$$

The melt of the fuse is after 0.2 seconds but at what current of the load, this can be found from the fuse curve.

Conversion table - American Wire Gauge - mm. - mm²

AWG N°	Diam. mm.	Area mm²		AWG N°	Diam. mm.	Area mm²
1	7,350	42,400		16	1,290	1,3100
2	6,540	33,600		17	1,150	1,0400
3	5,830	26,700		18	1,024	0,8230
4	5,190	21,200		19	0,912	0,6530
5	4,620	16,800		20	0,812	0,5190
6	4,110	13,300		21	0,723	0,4120
7	3,670	10,600		22	0,644	0,3250
8	3,260	8,350		23	0,573	0,2590
9	2,910	6,620		24	0,511	0,2050
10	2,590	5,270		25	0,455	0,1630
11	2,300	4,150		26	0,405	0,1280
12	2,050	3,310		27	0,361	0,1020
13	1,830	2,630		28	0,321	0,0804
14	1,630	2,080		29	0,286	0,0646
15	1,450	1,650		30	0,255	0,0503

From characteristics of the fuse for 0.2s and 30A normal rating the melting point at (copper melt at 1083 C°) short circuit is at around 270A.



Time/current characteristics of semi-enclosed fuses to BS 3036